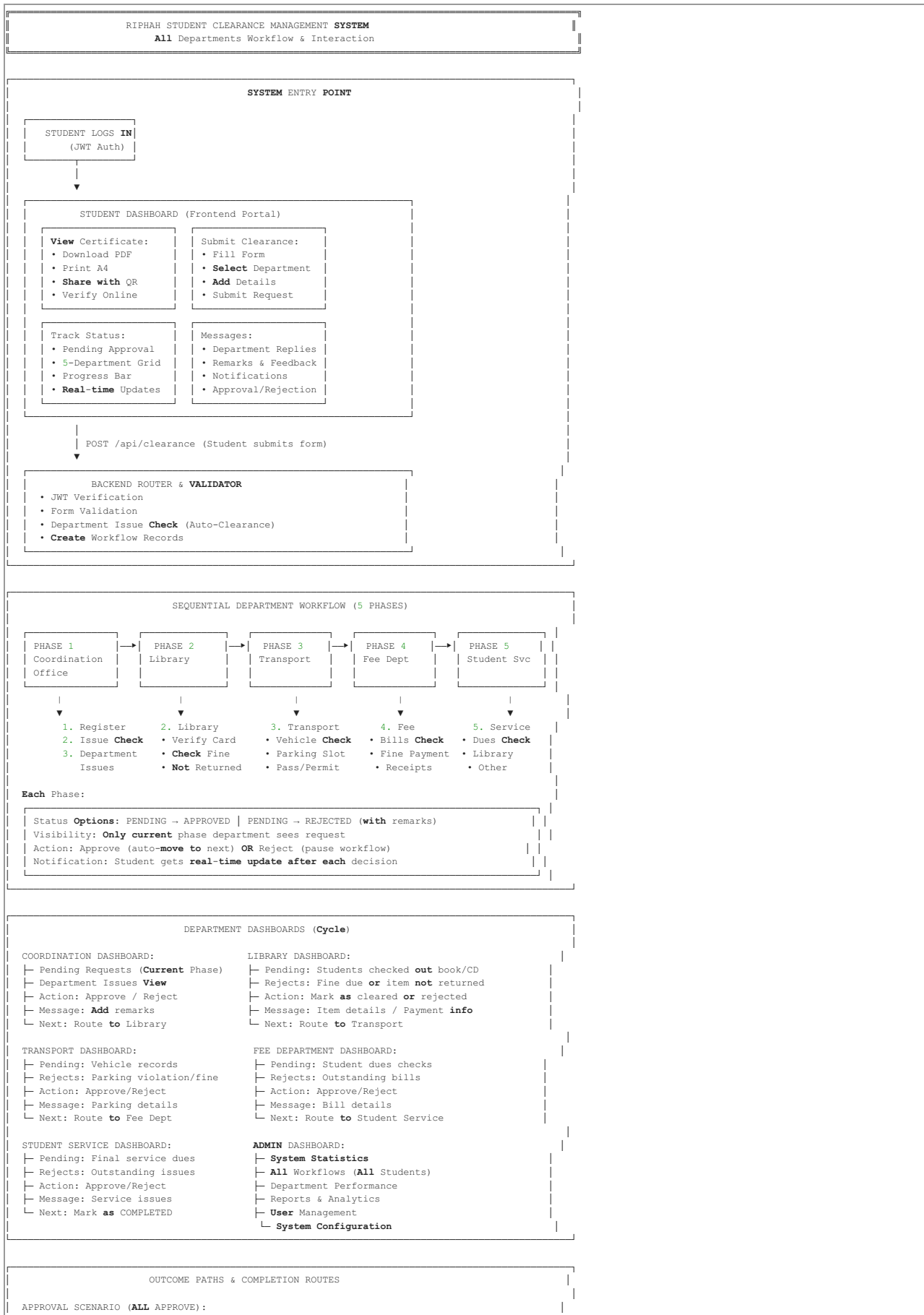


Complete Department Working Visualization

System Overview Diagram





SYSTEM ADMIN SUPERVISION & CONTROL

Admin Panel Features:

View ALL workflows (across all students)

Department Performance Metrics:

- Average approval time per department
- Rejection rate
- Pending request count
- Completion statistics

User Management:

- Create/Edit department staff
- Assign roles (coordination, library, transport, fee, service)
- View activity logs

System Configuration:

- Department settings
- Email templates
- Notification preferences

Reports:

- Monthly clearance trends
- Department-wise breakdown
- Bottleneck identification
- Student demographic analysis

Intervention:

- Can approve/reject (override if needed)
- Can reset workflow
- Can send bulk messages
- Can generate reports

Detailed Department Responsibilities

📅 Phase 1: Coordination Office (First Gate)

ROLE: Initial Registration & Department Issue Verification

Department Code: "Coordination"

Database Table: DepartmentIssue (coordination field)

Role Value: "coordination"

RESPONSIBILITIES:

- Verify student registration is complete
- Check coordination office has no pending issues
- Examples: Missing documents, incomplete registration, etc.

DATABASE CHECKS:

```
SELECT * FROM department_issues
WHERE student_id = ? AND department = 'coordination' AND status = ?
```

DECISION:

- ✓ APPROVED: No issues found → Route to Library
- ✗ REJECTED: Issues found → Workflow paused, notify student

DATA STORED IN: ClearanceWorkflow.phases[0]

Status: "Approved" | "Rejected"

Remarks: "All documents verified" | "Missing transcript"

📅 Phase 2: Library Department (Second Gate)

ROLE: Library Clearance & Returned Books Verification

Department Code: "Library"

Database Table: DepartmentIssue (library field)

Role Value: "library"

RESPONSIBILITIES:

- Check if student has library card registered
- Verify all books/CDs have been returned
- Check for outstanding fines
- Confirm library account is in good standing

DATABASE CHECKS:

SELECT * FROM library_records

WHERE student_id = ?

- Book status: "checked_out" (FAIL) vs "returned" (PASS)
- Fine amount: > 0 (FAIL) vs = 0 (PASS)
- Card status: active

DECISION:

✓ APPROVED: All books returned, no fines → Route to Transport

✗ REJECTED: Unreturned items or fines → Workflow paused

DATA STORED IN: ClearanceWorkflow.phases[1]

Status: "Approved" | "Rejected"

Remarks: "All items cleared" | "Book ID 12345 not returned + Rs. 500"

Phase 3: Transport Department (Third Gate)

ROLE: Vehicle & Parking Clearance

Department Code: "Transport"

Database Table: DepartmentIssue (transport field)

Role Value: "transport"

RESPONSIBILITIES:

- Verify parking permit status
- Check for traffic violations or parking fines
- Confirm vehicle is not impounded
- Verify shuttle/transport account clearance

DATABASE CHECKS:

SELECT * FROM transport_records

WHERE student_id = ?

- Vehicle status: "parked_legally" (PASS) vs "violation" (FAIL)
- Outstanding fines: Rs. 0 (PASS) vs > 0 (FAIL)
- Permit: "active" and not expired

DECISION:

✓ APPROVED: No violations, no fines, permit active → Route to Fee

✗ REJECTED: Violations or fines found → Workflow paused

DATA STORED IN: ClearanceWorkflow.phases[2]

Status: "Approved" | "Rejected"

Remarks: "Permit valid until 2026" | "Parking fine Rs. 1000 pending"

Phase 4: Fee Department (Fourth Gate)

ROLE: Financial Clearance & Dues Verification

Department Code: "Fee Department"

Database Table: DepartmentIssue (fee_department field)

Role Value: "feedeptment"

RESPONSIBILITIES:

- Verify tuition fees paid in full
- Check for outstanding hostel dues (if applicable)
- Verify lab/practical fees cleared
- Confirm all financial obligations settled
- Check for any pending payments

DATABASE CHECKS:

SELECT * FROM fee_records

WHERE student_id = ? AND for_year = ?

- Tuition paid: full amount received
- Lab_fees: Rs. 0 outstanding
- Hostel_dues: Rs. 0 outstanding
- Other_charges: all settled

DECISION:

✓ APPROVED: All fees paid → Route to Student Service

✗ REJECTED: Outstanding dues exist → Workflow paused

DATA STORED IN: ClearanceWorkflow.phases[3]

Status: "Approved" | "Rejected"

Remarks: "All fees cleared" | "Hostel dues Rs. 25,000 pending"

Phase 5: Student Service Department (Final Gate)

ROLE: Student Services Final Clearance

Department Code: "Student Service"

Database Table: DepartmentIssue (student_service field)

Role Value: "studentservice"

RESPONSIBILITIES:

- Verify student conduct **record** (no disciplinary action)
- **Check for** outstanding library/lab keys
- Verify **all** borrowed items returned
- Confirm **no** pending grievances
- Final clearance authority **before** certificate generation

DATABASE CHECKS:

SELECT * FROM student_services

WHERE student_id = ?

- Conduct_record: "good" (PASS) vs "violation" (FAIL)
- Outstanding_keys: 0 (PASS) vs > 0 (FAIL)
- Grievances: resolved (PASS) vs pending (FAIL)
- Borrowed_items: **all** returned

DECISION:

✓ APPROVED: **All** cleared → WORKFLOW COMPLETE

✗ REJECTED: Issues **found** → Workflow paused

ON APPROVAL:

1. **Set** status = "COMPLETED"

2. Generate certificate **with** QR code

3. Send email **with** PDF attachment

4. **Create system** notification

5. **Update** student dashboard

DATA STORED **IN**: ClearanceWorkflow.phases[4]

Status: "Approved" | "Rejected"

Remarks: "Ready for graduation" | "Lab key not returned"

 System Admin Role (Oversight & Control)

ROLE: System Administration & Monitoring

Database Table: User (where role = "admin")

Role Value: "admin"

DASHBOARD FEATURES:

1. VIEW ALL WORKFLOWS

- **Filter by**: Student **name**, SAP ID, status, **date** range
- **Show**: **All 5** phases **for each** student
- Timeline: Submission **date** → **Current** phase → Completion

2. DEPARTMENT STATISTICS

- Coordination: **45** pending, **1200** approved, **5** rejected
- Library: **32** pending, **950** approved, **12** rejected
- Transport: **28** pending, **1100** approved, **2** rejected
- Fee: **55** pending, **1050** approved, **8** rejected
- Student Service: **20** pending, **1180** approved, **3** rejected
- Average **time** per phase: **3-5** hours

3. PERFORMANCE METRICS

- Approval rate **by** department
- Average processing **time** per phase
- Bottleneck identification
- Peak request times
- Rejection rate trends

4. USER MANAGEMENT

- **Add/Edit/Delete** staff members
- Assign departments **and** roles
- **View** activity logs (who did what **and** when)
- **Reset** passwords **if** needed
- Track **login** history

5. SYSTEM OVERRIDE ACTIONS

- Manual approve/reject **any** phase (**with** reason)
- **Reset** student workflow **to start over**
- **Reset** specific phase **only**
- Send bulk notifications
- Manually generate certificate **if** needed

6. REPORTS & ANALYTICS

- Monthly clearance rate: **92%** completion
- Average days **to** complete: **8** days
- Department comparison: Which processes fastest?
- Student demographics: Semester-wise breakdown
- Export **to** Excel/PDF **for** meetings

7. CONFIGURATION

- Department names **and** codes
- Email templates **for** notifications
- Phase **order** (can reorder **if** needed)
- Certificate **template** customization
- **System** settings **and** preferences

Data Flow Architecture

DATA COLLECTIONS & FLOW

STUDENT SUBMISSION (Frontend -> Backend)

Student fills form:

- Student Name, SAP ID
- Father Name, Program, Semester
- Degree Status, Registration Number

POST /api/clearance (with JWT token)

Backend creates:

- ClearanceWorkflow (main workflow record)
- Phases array (5 stages initialized as PENDING)
- ComprehensiveClearanceValidation (tracking)

Check DepartmentIssue records

Auto-check logic:

FOR EACH department (Coordination -> Library -> Transport -> Fee -> Service):

```
SELECT * FROM department_issues
WHERE student_sapid = ? AND status = 'pending'
IF found: Set phase status = "Rejected"
ELSE: Set phase status = "Approved"
```

If ALL approved: Status = COMPLETED
If ANY rejected: Status = REJECTED

Generate Certificate:

- QR code with workflow ID
- PDF certificate (PDFKit)

Send Notifications:

- Email with PDF to student
- Messages to all department heads
- System notifications in dashboard

DEPARTMENT REVIEW (Cycle through each)

Department Staff logs in -> See requests for CURRENT PHASE ONLY

GET /api/clearance/department
Filter: overallStatus = "In Progress" AND currentPhase = dept_index

Staff reviews request details:

- Student name, SAP ID, program
- Department-specific info (books, fines, permits, fees, conduct)

Staff decision:

PUT /api/clearance/:workflowId/approve (with remarks)
or
PUT /api/clearance/:workflowId/reject (with reason)

Update workflow:

IF APPROVE:
• phases[currentPhase].status = "Approved"
• phases[currentPhase].approverName = staff_name
• currentPhase++ (move to next)
• IF currentPhase == 5: overallStatus = "Completed"
• ELSE: overallStatus = "In Progress"

IF REJECT:

- phases[currentPhase].status = "Rejected"
- overallStatus = "Rejected"
- Workflow paused (student can resubmit)

Send notifications:

- Student gets immediate update
- Next department notified (if approved)
- System logs activity

CERTIFICATE GENERATION & DELIVERY

Trigger: All phases approved (status = "COMPLETED")

Step 1: Generate Certificate

- Call certificateGenerator.js
- Input: Student data + approval info
- Output: PDF file (stored in /certificates folder)
- Include: QR code linking to /verify endpoint

Step 2: Send Email

- Call emailService.js
- To: student_email (from User record)
- Subject: "Clearance Approved - Certificate Attached"
- Attachment: PDF file
- Template: Professional HTML with university branding

Step 3: Create Message

- Save to Message collection
- Type: "notification"
- Show in student dashboard
- Include download link

Step 4: Update Records

- ClearanceWorkflow: completedAt = now
- ComprehensiveClearanceValidation: certificateGenerated = true
- Update dashboard refresh

Available for Student:

- View certificate in dashboard

- Download PDF file
- Print **to** A4 format
- Share **via** QR code
- **Public** verification at /verify/:id

Request State Lifecycle

STUDENT VIEW	DEPARTMENT VIEW	System State
[Submit Form]		
↓		
[Pending Approval] →	Coord: [Pending Request] ↓ (Reviews) [Approve]	Phase 0: Pending
↓		
[Library Checking] →	Lib: [Pending Request] ↓ (Reviews) [Approve]	Phase 1: Pending
↓		
[Transport Checking] →	Trans: [Pending Request] ↓ (Reviews) [Approve]	Phase 2: Pending
↓		
[Fee Verification] →	Fee: [Pending Request] ↓ (Reviews) [Approve]	Phase 3: Pending
↓		
[Service Final Check] →	Serv: [Pending Request] ↓ (Reviews) [Approve]	Phase 4: Pending
↓		
☑ APPROVED		Status: COMPLETED
↓		
[Certificate Ready]		
↓		
✉ Email Sent		
↓		
📄 Download		
↓		
[Cleared!]		
REJECTION PATH:		
[Submit Form]		
↓		
[Pending Approval] →	Coord: [Pending Request] ↓ (Reviews) [Approve]	Phase 0: Pending
↓		
[Library Checking] →	Lib: [Pending Request] ↓ (Reviews) [REJECT - Book not returned]	Phase 1: Pending
↓		
✖ REJECTED		Status: REJECTED
↓		
⚠ [Workflow Paused]		
↓		
🗉 [Message from Library]		
↓		
"Please return book ID 12345 by MM/DD/YY"		
↓		
Student resolves issue		
↓		
[Resubmit Request]		
↓		
[Library Re-checking] →	Lib: [Re-check Request] ↓ (Reviews again) [Approve]	Phase 1: Pending (Retry)
↓		
[Transport Checking] →	Trans: [Pending Request]	Phase 2: Pending
↓		
(Continue from Phase 2)		

Real-World Scenario Example

STUDENT: Ahmed Ali (SAP: 2019-0456)
TIMELINE:
Monday 10:00 AM - Student submits clearance request
└─ Auto-check runs:
├─ Coordination: ✓ No issues
├─ Library: ✗ Book not returned + Rs. 500 fine
├─ Transport: ✓ No issues
├─ Fee: ✓ No dues
└─ Student Service: ✓ No issues
└─ Result: REJECTED at Library phase
└─ Status: Workflow paused
└─ Student notification: "Please return book ID 2023-456 to library"
Monday 12:30 PM - Student goes to library
└─ Returns book
└─ Pays Rs. 500 fine
└─ Gets clearance from librarian
Monday 1:00 PM - Student clicks "RESUBMIT" in dashboard
└─ System re-checks library issues
└─ Library status now: ✓ CLEARED
└─ Auto-approval system continues from Transport
└─ All remaining departments approve automatically
├─ Transport: ✓ Approved
├─ Fee: ✓ Approved
└─ Student Service: ✓ Approved
└─ Status: COMPLETED
└─ Certificate generated
Monday 1:05 PM - Student receives email
└─ Subject: "Your Clearance is Approved!"
└─ Attachment: Clearance_Certificate_2019-0456.pdf
└─ Body: "Congratulations! You are cleared for graduation."
└─ Link: Download certificate / Verify online
Monday 2:00 PM - Student downloads certificate
└─ Opens PDF in dashboard
└─ Certificate shows:
├─ Name: Ahmed Ali
├─ SAP ID: 2019-0456
├─ All 5 departments: ✓ APPROVED
├─ Issue Date: MM/DD/2026
└─ QR Code: Verification link
└─ Prints A4 copy for records
Admin Dashboard View (Same time):
└─ New completion: Ahmed Ali (SAP 2019-0456)
└─ Total time: 7 hours (with resubmit)
└─ Department breakdown:
├─ Coordination: Passed immediately
├─ Library: Failed once, passed on resubmit
├─ Transport: Passed immediately
├─ Fee: Passed immediately
└─ Student Service: Passed immediately
└─ Certificate generated successfully

API Endpoints Quick Reference

STUDENT ENDPOINTS:		
POST	/api/clearance	Submit clearance request
GET	/api/clearance/student	View my clearance status
GET	/api/certificates	List my certificates
GET	/api/certificates/:id/download	Download certificate PDF
POST	/api/clearance/:id/resubmit	Resubmit rejected request
DEPARTMENT ENDPOINTS:		
GET	/api/clearance/department	View requests for my phase
PUT	/api/clearance/:id/approve	Approve a request
PUT	/api/clearance/:id/reject	Reject a request
GET	/api/clearance/department/history	View processed requests
ADMIN ENDPOINTS:		
GET	/api/admin/workflows	View all workflows
GET	/api/admin/statistics	Get department stats
POST	/api/admin/users	Manage staff users
GET	/api/admin/reports	Generate reports
PUT	/api/admin/override/:id	Manual override
PUBLIC ENDPOINT:		
GET	/api/verify/:certificateId	Verify certificate (public)

Summary Table

Department	Role	Input	Process	Output	Next Phase
Coordination	1st Gate	Student request	Verify registration	✓ Approve / ✗ Reject	Library
Library	2nd Gate	Books status	Check returned items & fines	✓ Approve / ✗ Reject	Transport
Transport	3rd Gate	Vehicle records	Check violations & permits	✓ Approve / ✗ Reject	Fee
Fee Department	4th Gate	Financial records	Check tuition & dues	✓ Approve / ✗ Reject	Student Service
Student Service	5th Gate	Conduct record	Final clearance	✓ Approve / ✗ Reject	Certificate
System Admin	Oversight	All data	Monitor & control	Reports & Metrics	Dashboard

Status: System is fully operational with all 7 roles integrated ☒

Architecture: Sequential multi-phase workflow with rejection handling ☒

Real-time: All updates visible to students immediately ☒